AP PRECALCULUS

Unit 1 • Worksheet 1.11B • Answer key

Worksheet 1.11B • Rational Functions and Concept of Limit

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the remainder when $x^3 - 2x + 5$ is divided by $x - 2$?

(A) 1
(B) 5
(C) 9
(D) 13

Solution

Answer: (C). $p(2) = 9$.

Question 2

NO CALCULATOR

What is the coefficient of x^3 in $(x+2)^5$?

(A) 10
(B) 20
(C) 80
(D) 40

Solution

Answer: (D). $10 \cdot 2^2 = 40$.

Question 3

NO CALCULATOR

Which is the slant asymptote of $r(x) = \dfrac{2x^2 + 3x - 4}{x + 2}$?

(A) $y = 2x - 1$
(B) $y = 2x + 7$
(C) $y = 2x + 3$
(D) $y = x - 1$

Solution

Answer: (A). Synthetic division gives $2x - 1$, remainder -2.

Question 4 (Free response) NO CALCULATOR

Let $p(x) = 2x^3 + x^2 - 13x + 6$.

(a) Show that $x = 2$ is a zero, and factor p completely.

(b) Write $\dfrac{p(x)}{x+1}$ in the form $q(x) + \dfrac{k}{x+1}$.

(c) Use (b) to find $\lim\limits_{x \to -1^+} \dfrac{p(x)}{x+1}$.

Solution

(a) $p(2) = 0$; $p = (x-2)(2x-1)(x+3)$.
(b) $2x^2 - x - 12 + \frac{18}{x+1}$.
(c) $\frac{18}{x+1} \to +\infty$: the limit is $+\infty$.

Question 5 (Free response) NO CALCULATOR

Expand: (a) $(x-2)^4$ (b) $(2x+1)^3$ (c) $(x+y)^5$.

Solution

(a) $x^4 - 8x^3 + 24x^2 - 32x + 16$. (b) $8x^3 + 12x^2 + 6x + 1$. (c) $x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$.